The General Theory of Thermohydraulic Equivalence

A Conditional Framework for Comparing Heat-Transfer Fluids

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Abstract

This treatise develops a general framework for determining when two coolant formulations may be treated as thermohydraulically equivalent within a specified engineering service. The central proposition is that equivalence is not an intrinsic label attached to a fluid independently of its application. It is a conditional relation defined over a declared operating domain, apparatus, control constraint, system boundary, response vector, uncertainty model, and tolerance structure. The framework couples thermodynamic heat transport, convective heat transfer, fluid mechanics, pump energetics, operating-envelope constraints, and statistical uncertainty. It distinguishes property similarity from component-level and system-level performance equivalence; derives dimensionless representations; formalises exact and tolerance-based equivalence; explains why pairwise tolerance similarity is generally non-transitive; and proposes robust, uncertainty-aware decision rules. A reproducible computational programme is outlined using manufacturer-reported property observations and transparent pipe-flow screening models. Such models can generate testable hypotheses but cannot substitute for harmonised measurements or controlled experiments. The theory therefore ends with an experimental protocol and a hierarchy of claims, from property correspondence to validated service substitution.

The treastise ends with "The End"

Keywords: coolant; thermohydraulics; heat transfer; hydraulic resistance; Reynolds number; Prandtl number; Nusselt number; equivalence relation; uncertainty quantification; fluid substitution.

1 Introduction

Engineering comparisons of coolants are often reduced to a small number of tabulated properties: specific heat capacity, thermal conductivity, density, and viscosity. These quantities are indispensable, but no one of them independently determines the performance of a cooling system. A fluid with high thermal conductivity may impose a different pumping burden; a fluid with high specific heat may have a comparatively high viscosity; a dense fluid may alter both volumetric flow and hydraulic losses. The relevant trade-off depends on the geometry, operating constraint, heat load, temperature field, pressure boundary conditions, pump, heat exchanger, and material system.

The question is consequently not whether two fluids are universally identical. It is whether they deliver sufficiently similar service under a clearly defined set of conditions. This is the problem of thermohydraulic equivalence.

The theory developed here has five objectives:

1. define the comparison object and its operating domain;
2. derive the coupled thermal and hydraulic response of a coolant system;
3. distinguish several levels of equivalence;
4. construct rigorous tolerance- and uncertainty-aware decision rules; and
5. specify a reproducible route from manufacturer data and simulation to experimental validation.

The framework is deliberately conditional. A conclusion obtained for one pipe diameter, flow constraint, and temperature range must not be silently transferred to a different cold plate, heat exchanger, pump curve, or application.

2 The comparison object and operating domain

Let a coolant formulation be denoted by F. Its state-dependent thermophysical vector is

$$\mathbf{x}_F(T,p,c) = \begin{bmatrix} c_p(T,p,c) \\ k(T,p,c) \\ \rho(T,p,c) \\ \mu(T,p,c) \end{bmatrix}, \tag{1}$$

where T is temperature, p is absolute pressure, c denotes composition or formulation state, c_p is specific heat capacity, k is thermal conductivity, ρ is density, and μ is dynamic viscosity.

The formulation identity must include the commercial grade, concentration, concentration basis, additive package where disclosed, and relevant preparation procedure. A grade suffix is not a concentration measurement. A mass fraction and a volume fraction are not interchangeable without a density-aware conversion.

Let the operating domain be

$$\Omega = \{(T,p,\dot{m},\mathbf{b},\mathbf{u}) : T \in \mathcal{T},\ p \in \mathcal{P},\ \dot{m} \in \mathcal{M},\ \mathbf{b} \in \mathcal{B},\ \mathbf{u} \in \mathcal{U}\}, \tag{2}$$

where $\dot{m}$ is mass flow, $\mathbf{b}$ collects boundary conditions, and $\mathbf{u}$ collects controllable operating variables. The domain must also specify geometry, surface roughness, orientation, wall material, exchanger configuration, and system boundary.

A comparison is incomplete unless it states what is held fixed. At minimum, the following constraints should be treated as distinct experiments:

Equal mass flow: $\dot{m}_1 = \dot{m}_2$.

Equal volumetric flow: $\dot{V}_1 = \dot{V}_2$.

Equal pump input: $P_{\mathrm{el},1} = P_{\mathrm{el},2}$.

Equal delivered thermal service: the fluids are operated to satisfy the same required heat duty and temperature constraints, while the required pumping and auxiliary power are measured.

These constraints are not equivalent. In particular, equal volumetric flow does not imply equal mass flow when densities differ.

3 Thermodynamic heat transport

For a steady flowing stream, the enthalpy transport rate is

$$\dot{Q}_{\mathrm{fluid}} = \dot{m}\,(h_{\mathrm{out}} - h_{\mathrm{in}})\,. \tag{3}$$

For single-phase sensible heating with approximately constant specific heat,

$$\dot{Q}_{\mathrm{fluid}} \simeq \dot{m} c_p\,(T_{\mathrm{out}} - T_{\mathrm{in}})\,. \tag{4}$$

The enthalpy expression is more general and should be preferred when heat capacity varies materially with temperature, composition changes, or phase change occurs.

For a control volume with stored energy and heat leakage, the balance is more properly written

$$\dot{Q}_{\mathrm{in}} - \dot{Q}_{\mathrm{out}} - \dot{Q}_{\mathrm{loss}} = \frac{dE_{\mathrm{stored}}}{dt}. \tag{5}$$

Consequently, a mismatch between heater input and coolant enthalpy gain is not automatically evidence of a defective property model. It may reflect thermal storage, parasitic heat transfer, sensor bias, flowmeter error, or an incorrectly specified control volume.

For a heat exchanger under the assumptions appropriate to the selected configuration,

$$\dot{Q} = UA\Delta T_{\mathrm{lm}}, \tag{6}$$

where U is the overall heat-transfer coefficient, A is the reference area, and ΔT_{lm} is the logarithmic mean temperature difference. The overall resistance includes fluid-side convection, wall conduction, fouling, and any additional layers. In a simplified planar resistance representation,

$$\frac{1}{U} = \frac{1}{h_i} + R_{\mathrm{wall}} + R_{\mathrm{fouling}} + \frac{1}{h_o}, \tag{7}$$

provided all resistances are expressed on a consistent area basis. For cylindrical walls, the wall resistance must be expressed using the appropriate logarithmic radial geometry.

4 Hydraulic response and pumping energetics

For fully developed pipe flow, the Darcy–Weisbach relation gives

$$\Delta P = f_D \frac{L}{D} \frac{\rho v^2}{2}, \tag{8}$$

where f_D is the Darcy friction factor, L is pipe length, D is internal diameter, and v is mean velocity. For a circular pipe,

$$v = \frac{4\dot{m}}{\pi \rho D^2}, \qquad \dot{V} = \frac{\dot{m}}{\rho}. \tag{9}$$

Substitution yields

$$\Delta P = \frac{8 f_D L \dot{m}^2}{\pi^2 \rho D^5}. \tag{10}$$

The hydraulic power transferred to the fluid is

$$P_{\mathrm{hyd}} = \Delta P \dot{V}, \tag{11}$$

and the electrical input to a pump is approximately

$$P_{\mathrm{el,pump}} = \frac{\Delta P \dot{V}}{\eta_p}, \tag{12}$$

where η_p is the pump-and-drive efficiency appropriate to the defined boundary.

The familiar cubic dependence of pumping power on flow is a restricted approximation. It requires simplifying assumptions about friction factor, density, geometry, and efficiency. In actual operation, friction factor varies with Reynolds number and relative roughness, while pump efficiency varies with operating point. The system curve and pump curve jointly determine the realised flow.

For a complete circuit, local losses must also be included:

$$\Delta P_{\mathrm{system}} = \Delta P_{\mathrm{distributed}} + \sum_j K_j \frac{\rho v_j^2}{2} + \Delta P_{\mathrm{components}}, \tag{13}$$

where the summation represents fittings and other local restrictions, and the final term may include cold plates, manifolds, filters, valves, and heat exchangers not already represented by the local-loss coefficients.

5 Dimensionless structure of convective heat transfer

The principal dimensionless groups for forced convection are

$$Re = \frac{\rho v D}{\mu}, \tag{14}$$

$$Pr = \frac{\mu c_p}{k}, \tag{15}$$

$$Nu = \frac{h D}{k}. \tag{16}$$

A heat-transfer correlation has the generic form

$$Nu = \Phi(Re, Pr, \varepsilon/D, \mathrm{geometry}, \mathrm{boundary\ condition}, \ldots), \tag{17}$$

where ε is absolute roughness. A correlation is admissible only within its stated flow regime, geometry, property-evaluation convention, and validity range.

For example, the Dittus–Boelter expression is commonly written

$$Nu = 0.023 Re^{0.8} Pr^{n}, \tag{18}$$

with the exponent selected according to the heating or cooling convention. It is not a universal law. The Gnielinski relation is often more appropriate over a broader turbulent pipe-flow range, while laminar and thermally developing flows require different treatments.

Under the illustrative power-law approximation $Nu = CRe^{m}Pr^{n}$, with fixed geometry and mean velocity, the implied dependence of h is

$$h \propto \rho^{m} \mu^{n-m} c_p^{n} k^{1-n}. \tag{19}$$

This expression is useful for understanding competing property effects. It is not a universal coolant ranking formula: the exponents, admissible regime, and fixed operating variables depend on the chosen correlation and experiment.

6 A hierarchy of equivalence

The word equivalence must be qualified by the response being compared. We define the following hierarchy.

Property equivalence. Selected entries of $\mathbf{x}_F$ agree within declared tolerances over a specified state domain. This is a statement about properties, not system performance.

Thermal equivalence. The fluids deliver equivalent thermal response under the same specified thermal boundary conditions. The response may include heat duty, outlet temperature, wall temperature, and thermal transient behaviour.

Hydraulic equivalence. The fluids produce equivalent hydraulic response under a specified flow or control constraint. Relevant responses include pressure drop, flow rate, pump input, and operating stability.

Thermohydraulic equivalence. The fluids jointly satisfy declared thermal and hydraulic response tolerances under the same apparatus and operating protocol.

System-service equivalence. The complete systems deliver the same required service, including thermal limits and reliability constraints, while differences in electrical energy, auxiliary loads, maintenance, and lifecycle costs are explicitly reported.

These levels must not be conflated. Similar specific heat capacities do not establish thermal equivalence. Similar thermal duty does not establish hydraulic equivalence. Thermohydraulic equivalence does not establish chemical compatibility, electrical insulation, corrosion resistance, toxicity equivalence, flammability equivalence, environmental equivalence, or lifecycle-cost equivalence.

7 Formal definition of thermohydraulic equivalence

Let F denote a fluid formulation and let $\mathcal{A}$ denote a fully specified apparatus and protocol. Define the response map

$$\mathbf{y}_{F,\mathcal{A}}(\boldsymbol{\xi}) = \begin{bmatrix} \dot{Q} \\ \Delta P \\ P_{\mathrm{el,pump}} \\ T_{\mathrm{wall,max}} \\ T_{\mathrm{out}} \\ s_1 \\ \vdots \\ s_r \end{bmatrix}, \tag{20}$$

where $\boldsymbol{\xi} \in \Omega$ collects the operating conditions and s_j are declared stability or operational metrics. The system boundary and definitions of every component of $\mathbf{y}$ must be fixed before comparison.

Two fluids F_1 and F_2 are exactly thermohydraulically equivalent on $(\Omega, \mathcal{A})$ if

$$\mathbf{y}_{F_1,\mathcal{A}}(\boldsymbol{\xi}) = \mathbf{y}_{F_2,\mathcal{A}}(\boldsymbol{\xi}) \quad \text{for every } \boldsymbol{\xi} \in \Omega, \tag{21}$$

and both satisfy the same admissibility constraints throughout the domain.

Exact equality is rarely a practical engineering requirement. Let $\mathbf{W}$ be a diagonal or positive-definite scaling matrix, and let $\boldsymbol{\tau}$ be a vector of tolerances. A pointwise tolerance rule is

$$|\mathbf{W}\left[\mathbf{y}_{F_1,\mathcal{A}}(\boldsymbol{\xi}) - \mathbf{y}_{F_2,\mathcal{A}}(\boldsymbol{\xi})\right]| \leq \boldsymbol{\tau}. \tag{22}$$

The scaling matrix is used to make units and magnitudes commensurable; it must not conceal the original physical units or the engineering meaning of the tolerances. A more general norm-based rule is

$$d_{\mathcal{A}}(F_1, F_2; \Omega) = \sup_{\boldsymbol{\xi} \in \Omega} \left\|\mathbf{W}\left(\mathbf{y}_{F_1,\mathcal{A}}(\boldsymbol{\xi}) - \mathbf{y}_{F_2,\mathcal{A}}(\boldsymbol{\xi})\right)\right\|_{\mathcal{N}} \leq \tau, \tag{23}$$

where the norm $\mathcal{N}$, weighting, and scalar tolerance τ are declared in advance.

The use of a supremum is conservative: it requires the tolerance to hold throughout the declared domain. If only a finite set of operating points is tested, the conclusion is restricted to those points unless interpolation between them is supported by a validated model and uncertainty bounds.

8 Why tolerance similarity is not automatically an equivalence relation

Mathematical equivalence relations are reflexive, symmetric, and transitive. Exact equality of response functions has these properties. A pairwise tolerance rule generally does not.

Consider three scalar responses $y_A = 0$, $y_B = 0.75\tau$, and $y_C = 1.5\tau$. Then A is within tolerance of B, and B is within tolerance of C, but A is not within tolerance of C. Therefore,

$$A \sim_\tau B, \quad B \sim_\tau C \quad \not\Rightarrow \quad A \sim_\tau C. \tag{24}$$

Consequently, a graph in which edges connect pairwise-near fluids is a similarity network, not necessarily a partition into mathematical equivalence classes. Connected components can chain together fluids whose endpoints are not mutually within tolerance.

There are three defensible approaches:

1. report pairwise similarity edges without calling connected components equivalence classes;
2. use complete-linkage clusters, requiring every pair in a cluster to satisfy the tolerance; or
3. define a canonical representative and require every member to satisfy a predeclared maximum deviation from that representative, while separately checking the cluster diameter.

The choice is a methodological decision and must be reported. A clustering algorithm cannot turn an arbitrary tolerance into a transitive relation without changing the definition.

9 Uncertainty-aware equivalence

Manufacturer data may be typical values, guaranteed limits, calculated values, or measurements. These evidence classes are not interchangeable. A property table without measurement uncertainty does not justify treating the listed value as exact.

Let $\widehat{\mathbf{y}}_F$ be an estimated response and let $\mathbf{U}_F$ describe its uncertainty, including property uncertainty, instrument uncertainty, model discrepancy, and parameter uncertainty. The difference between two estimated responses has uncertainty contributions from both fluids and their covariance. For a scalar response difference $\widehat{\delta}$, a simple expanded-uncertainty rule is

$$|\widehat{\delta}| + k_u u_\delta \leq \tau, \tag{25}$$

where u_δ is the combined standard uncertainty and k_u is a declared coverage factor. This is a conservative sufficient rule under the stated uncertainty approximation, not a universal statistical guarantee.

For vector responses, define an uncertainty region $\mathcal{C}_\delta$ for the response difference. A robust equivalence criterion requires

$$\mathcal{C}_\delta(\boldsymbol{\xi}) \subseteq \mathcal{T}_{\text{accept}} \quad \text{for all tested } \boldsymbol{\xi}, \tag{26}$$

where $\mathcal{T}_{\text{accept}}$ is the predeclared tolerance region. A probabilistic alternative may require

$$\Pr\left(\mathbf{y}_{F_1} - \mathbf{y}_{F_2} \in \mathcal{T}_{\text{accept}}\right) \geq 1 - \alpha. \tag{27}$$

The confidence or probability statement must specify the sampling model, uncertainty distribution, and treatment of model discrepancy.

Failure to reject a difference is not proof of equivalence. Equivalence testing requires a predeclared acceptance region and sufficient precision to show that the difference lies inside it.

10 Operating-envelope and feasibility constraints

A candidate fluid is not admissible merely because its predicted heat-transfer coefficient is high. Its full operating trajectory must remain inside a documented feasible region. Let

$$\mathcal{F}_F = \left\{ \boldsymbol{\xi} : \begin{array}{l} T_{\min,F} \leq T \leq T_{\max,F}, \\ p_{\min,F}(T,c) \leq p \leq p_{\max,F}(T,c), \\ \text{single-phase or approved phase regime,} \\ \text{materials and chemical constraints satisfied} \end{array} \right\}. \tag{28}$$

The operating domain must satisfy $\Omega \subseteq \mathcal{F}_F$, subject to the evidence and safety margins appropriate to the application.

For mixtures, phase behaviour depends on composition and pressure as well as temperature. A pure-fluid saturation curve is not a substitute for a mixture phase envelope. Two-phase systems require additional treatment of latent heat, vapour quality, dryout, flow instability, pressure oscillations, and condensation. The present single-phase pipe-flow screening model does not establish equivalence for two-phase cooling.

11 Multiobjective performance and economic interpretation

Thermal duty, pressure loss, pump electricity, maximum wall temperature, stability, and service life are not naturally reducible to one universal scalar. A candidate may improve one objective while worsening another. The appropriate first presentation is therefore a response vector and a Pareto analysis.

A simple screening ratio,

$$\mathcal{E}_{\text{screen}} = \frac{\dot{Q}}{P_{\text{el,pump}}}, \tag{29}$$

can be informative within a carefully defined pump-only boundary. It is not a complete coefficient of performance for a cooling plant. A full-system measure must include compressors, fans, cooling towers, controls, and other material auxiliary loads.

For lifecycle analysis, a discounted cost representation may be written

$$NPV_F = C_{0,F} + \sum_{t=1}^{N} \frac{C_{\text{energy},F,t} + C_{\text{maintenance},F,t} + C_{\text{replacement},F,t} + C_{\text{downtime},F,t} + C_{\text{disposal},F,t}}{(1+r)^t}, \tag{30}$$

where $C_{0,F}$ is initial capital and fluid cost, r is the discount rate, and N is the analysis horizon. The service delivered, energy-price path, replacement assumptions, and residual value must be made explicit. A lower pumping requirement does not automatically imply lower lifecycle cost if the fluid requires more frequent replacement or specialised materials.

12 Computational implementation and reproducibility

A reproducible implementation should preserve a strict distinction between observations and model outputs. Each property record should retain product identity, formulation and concentration basis, temperature, pressure basis, original units, converted units, source revision, page or table, evidence type, and uncertainty status.

A computational pipeline can proceed as follows:

1. validate product identifiers, source identifiers, units, and duplicate observations;
2. select only comparable formulations and documented state points;
3. interpolate only within observed temperature intervals when justified, preserving the original endpoints and labelling the result as interpolated;
4. calculate Re, Pr, friction factor, pressure drop, pump power, Nu, and h using correlations valid for the operating point;
5. flag points outside the selected correlation envelope rather than extrapolating silently;
6. evaluate each fluid under equal mass flow, equal volumetric flow, and equal pump input separately;
7. construct response surfaces over the common observed domain;
8. propagate input and model uncertainty; and
9. export machine-readable outputs, assumptions, source identifiers, and validation results.

The repository accompanying this treatise uses manufacturer-reported observations for an initial cohort and transparent single-phase pipe-flow models for screening. Its modelled HC-grade surfaces are conditional on illustrative pipe geometry, assumed roughness, pump efficiency, and selected correlations. The HC-10/20/30/40 property cohort is sourced from a 2025 engineering guide, whereas HC-50 remains sourced from a 2020 technical data sheet. This source-revision asymmetry is a limitation, not a detail to be hidden. The model outputs are hypotheses for experimental testing; they are not independent measurements or proof of product equivalence.

13 Experimental identification and validation

A controlled comparison should use the same test section, wetted materials, fittings, orientation, heat exchanger, pump, instrumentation, and control protocol. The test record should identify fluid lot, age, composition, dilution water, degassing procedure, cleaning procedure, and any carryover risk.

At minimum, measure:

- mass flow or calibrated volumetric flow and fluid density;
- matched inlet and outlet temperatures;
- wall temperatures at declared locations;
- differential pressure across the test section and relevant components;
- pump electrical input and, where possible, shaft or hydraulic power;
- heat input, heat loss, and ambient conditions;
- fluid composition or concentration before and after testing; and
- instrument calibration and uncertainty.

Run separate matched-condition experiments for equal mass flow, equal volumetric flow, and equal pump electrical input. Where service substitution is the objective, add a controlled equal-thermal-service experiment. Randomise test order where practicable, repeat each condition, and report calibration before and after the campaign. Long-duration tests should examine fouling, corrosion, degradation, seal compatibility, and changes in fluid properties.

Equivalence margins must be specified before inspecting the results. The analysis should distinguish repeatability, reproducibility, instrument uncertainty, and model discrepancy. A fluid pair may be thermohydraulically equivalent in a short-duration clean-loop test and still fail a lifecycle or materials-compatibility qualification.

14 Propositions and deductions

Proposition 1 (Conditionality). Thermohydraulic equivalence is defined relative to an apparatus, operating domain, control constraint, response vector, and tolerance structure.

Argument. The response map depends on fluid properties and on geometry, boundary conditions, pump characteristics, and control variables. Changing any of these arguments changes the function being compared. Thus an equivalence finding for one declared configuration does not logically imply equivalence under another configuration.

Proposition 2 (Property similarity is insufficient). Equality or similarity of a strict subset of thermophysical properties does not, by itself, imply thermohydraulic equivalence.

Argument. The thermal response depends on enthalpy transport and convective resistance, while hydraulic response depends on density, viscosity, flow, geometry, and friction. A subset of equal properties leaves other determinants unconstrained. Therefore the response vectors need not agree.

Proposition 3 (Constraint dependence). The relative performance ordering of two fluids can change when the comparison constraint changes.

Argument. At equal mass flow, volumetric flow is inversely proportional to density. At equal volumetric flow, mass flow is proportional to density. Pump power depends on pressure loss and volumetric flow, and pressure loss depends on the flow regime and friction factor. The two experiments therefore evaluate different operating points and need not preserve rankings.

Proposition 4 (Tolerance non-transitivity). A symmetric pairwise tolerance rule need not be transitive.

Argument. The scalar counterexample in Section 7 gives two adjacent differences smaller than the tolerance and an endpoint difference larger than the tolerance. Hence pairwise tolerance similarity is not generally an equivalence relation.

Proposition 5 (Evidence limitation). A deterministic simulation based on manufacturer typical values cannot establish empirical equivalence without uncertainty characterisation and experimental validation.

Argument. The simulation output is conditional on input values, constitutive correlations, and model assumptions. If input uncertainty and model discrepancy are unquantified, the calculated response difference is not a validated estimate of the physical difference. Independent measurements under a controlled protocol are required for an empirical equivalence claim.

15 Limitations and research agenda

The framework is general in structure, but each application requires validated constitutive models and data. The present repository seed is small and heterogeneous. Manufacturer data may differ in test method, pressure basis, reporting precision, and revision. Some concentration bases are unknown. The current HC-50 source is older than the other HC-grade source. No independent laboratory measurements are yet available for the initial cohort.

Further work should include:

1. harmonised property measurements over overlapping temperature and pressure domains;
2. uncertainty-aware property surfaces with physically admissible constraints;
3. validated laminar, transitional, turbulent, and developing-flow models;
4. complete exchanger and cold-plate models with wall and fouling resistances;
5. pump-curve coupling and dynamic control analysis;
6. phase-envelope and stability models for mixtures and two-phase fluids;
7. corrosion, compatibility, degradation, and contamination testing;
8. lifecycle and environmental assessment; and
9. preregistered equivalence margins with independent replication.

16 Conclusion

Thermohydraulic equivalence is not a universal identity between coolant products. It is a rigorously bounded engineering conclusion about the similarity of specified thermal and hydraulic responses under a declared apparatus, operating domain, control constraint, system boundary, and uncertainty model.

The theory separates property equivalence, thermal equivalence, hydraulic equivalence, thermohydraulic equivalence, and complete system-service equivalence. It demonstrates why a fluid ranking can depend on whether mass flow, volumetric flow, pump input, or delivered service is held constant. It also shows why pairwise tolerance similarity must not be mistaken for a transitive mathematical equivalence relation.

The practical consequence is methodological: every equivalence claim must state what was compared, where it was compared, under which constraints, with what tolerances, and with what uncertainty. Transparent computation can identify candidate substitutions and design experiments. Only harmonised evidence and controlled validation can establish that a proposed substitution is fit for service.

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The End